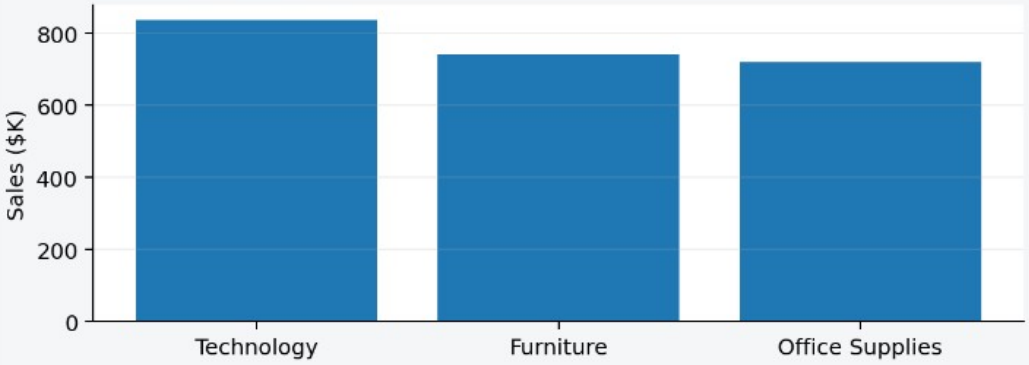
Name: Siddhant Chauhan **Lab Sheet 5** **Roll No: 56**

SUPERSTORE SALES — INTERACTIVE BUSINESS DASHBOARD

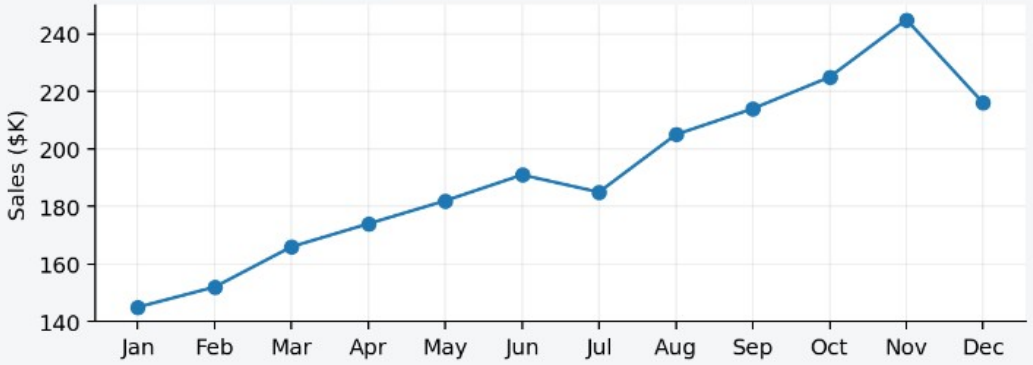
Filters: Region • Category • Year

Total Sales	Total Profit	Orders	Avg. Order Value
$2.30M	$286K	9,994	$230

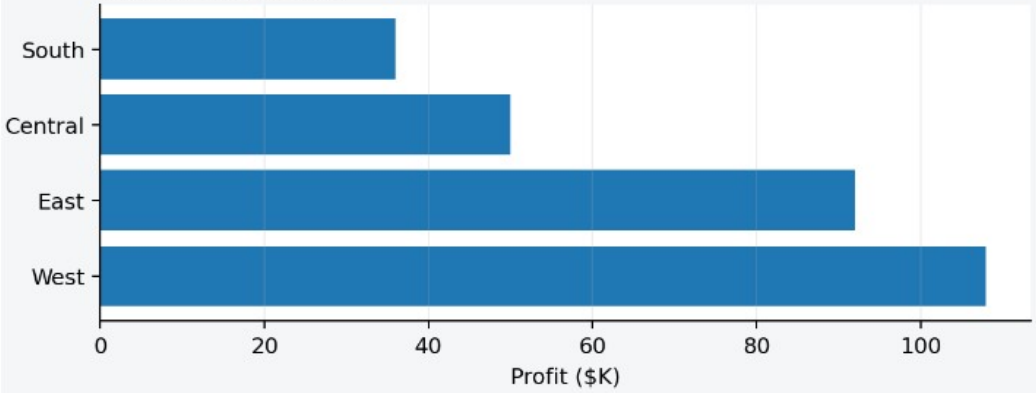
Sales by Category



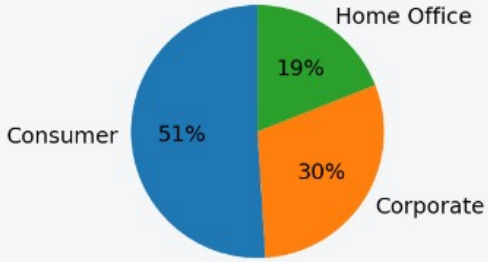
Monthly Sales Trend



Profit by Region



Sales by Customer Segment



Theory Questions

1. What is Business Intelligence (BI)? How does it support organizational decision making?

Business Intelligence (BI) is the process of collecting, analyzing, and presenting data to help organizations make better decisions. BI converts raw data into meaningful information using reports, dashboards, and visualizations. It helps managers identify trends, monitor performance, reduce risks, and make data-driven decisions.

2. What is an interactive dashboard? How is it different from a static report?

An interactive dashboard is a visual display of data where users can interact with charts, filters, slicers, and other elements. A static report mainly presents fixed information that does not change based on user interaction. Interactive dashboards allow users to explore data and get specific insights quickly.

3. Compare Tableau and Microsoft Power BI based on features, usability, and business applications.

Feature	Tableau	Power BI
Visualization	Very powerful and flexible	Strong and easy to use
Usability	Suitable for advanced users	Beginner-friendly
Integration	Supports many data sources	Excellent Microsoft integration
Cost	Generally more expensive	More affordable
Business Use	Advanced analytics and visualization	Reporting, dashboards, and business analytics

Both tools are widely used for data analysis, visualization, and business decision-making.

4. What are Key Performance Indicators (KPIs)? Give examples of KPIs used in sales analytics.

KPIs are measurable values used to evaluate the performance of a business or activity. They help organizations understand whether they are achieving their goals.

Examples of sales KPIs:

- Total Sales Revenue
- Sales Growth Rate
- Number of Orders
- Average Order Value
- Profit Margin

- Conversion Rate
- Customer Acquisition Cost

5. Explain the purpose of filters, slicers, and drill-down functionality in dashboards.

Filters are used to display only selected data based on specific conditions. **Slicers** provide a visual way to filter dashboard data, such as selecting a year, region, or product. **Drill-down** allows users to move from summarized information to more detailed information, such as Year → Quarter → Month.

6. What factors should be considered while designing an effective business dashboard?

An effective dashboard should consider:

- Clear and simple design
- Relevant KPIs and data
- Easy-to-understand charts
- Proper use of colors and labels
- Interactive filters and slicers
- Fast performance
- Target audience and business objectives

The dashboard should present important information without unnecessary complexity.

7. How can dashboards help managers monitor organizational performance in real time?

Dashboards can connect to regularly updated data sources and display the latest information. Managers can monitor sales, revenue, expenses, inventory, and employee performance from a single screen. Alerts and KPIs can also highlight problems quickly, helping managers take timely corrective actions.

8. Why is data visualization important in Business Intelligence tools?

Data visualization represents complex data using charts, graphs, maps, and dashboards. It makes large amounts of information easier to understand. Visualization helps users identify trends, patterns, relationships, and unusual values quickly, which improves decision-making.

9. What insights can be derived from a sales dashboard for an e-commerce company?

An e-commerce sales dashboard can provide insights such as:

- Total sales and profit
- Best-selling products
- Sales by region

- Monthly or yearly sales trends
- Number of orders
- Average order value
- Customer purchasing behavior
- Product returns and cancellations

These insights help the company improve sales and business strategies.

10. Explain how interactive dashboards improve communication between data analysts and business stakeholders.

Interactive dashboards provide a common visual platform for analysts and business stakeholders. Stakeholders can use filters, slicers, and drill-downs to explore the data themselves. This reduces confusion, makes insights easier to understand, and helps both sides discuss business problems using the same data.